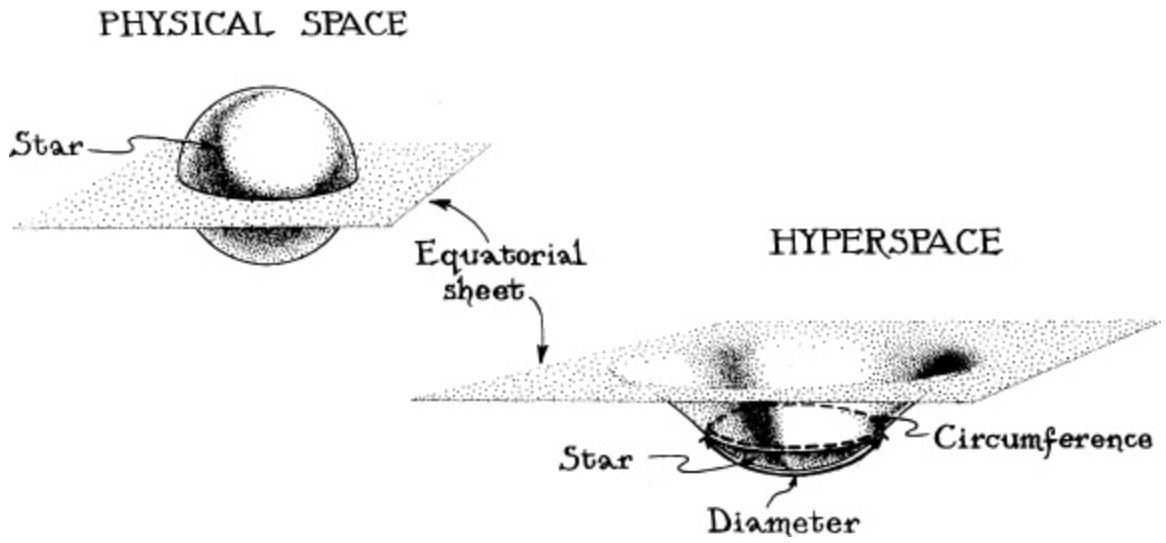
uncurved hyperspace, the sheet can maintain its curved geometry only by bending downward like a bowl. Such diagrams of two-dimensional sheets from our curved Universe, embedded in a hypothetical, flat, three-dimensional hyperspace, are called *embedding diagrams.*



3.3 The curvature of the three-dimensional space inside and around a star (upper left), as depicted by means of an *embedding diagram* (lower right). This is the curvature predicted by Schwarzschild's solution to Einstein's field equation.

It is tempting to think of hyperspace's third dimension as being the same as the third spatial dimension of our own Universe. We must avoid this temptation. Hyperspace's third dimension has nothing whatsoever to do with any of the dimensions of our own Universe. It is a dimension into which we can never go and never see, and from which we can never get any information; it is purely hypothetical. Nonetheless, it is useful. It helps us visualize the Schwarzschild geometry, and later in this book it will help us visualize other curved-space geometries: those of black holes, gravitational waves, singularities, and worm-holes (Chapters 6, 7, 10, 13, and 14).

As the embedding diagram in Figure 3.3 shows, the Schwarzschild geometry of the star's equatorial sheet is qualitatively the same as the geometry of the 2D beings' universe: Inside the star, the geometry is bowl-like and curved; far from the star it becomes flat. As with the large circle in the 2D beings' bowl (Figure 3.2), so also here (Figure 3.3), the star's circumference divided by its diameter is less than π. For our Sun, the ratio of circumference to diameter is predicted to be less than π by several parts in a million; in other words, inside the Sun, space is flat to within several